

Reducing the Environmental Footprints of Public Cloud Workloads through Spatial and Temporal Shifting: FaaS and Big Data Case Studies

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Abstract—In this paper, we investigate the potential of spatial and temporal shifting of cloud workloads reduce their environmental footprint. We conduct a simulation study using real-world data from multiple cloud providers (AWS and Azure) and workload traces (Spark and FaaS). Our results show that spatial shifting can lead to significant reductions in carbon, water, and land use footprints, with improvements ranging from 20% to 85%, depending on the scenario and optimization criteria. Also temporal shifting achieves footprints' reduction, but to a smaller scale. The combination of both methods have the highest reductions, although the incremental benefit over spatial shifting alone is small. We conducted sensitivity analyses to show that our approach is resilient to prediction errors in grid mix data and across different seasons.

Index Terms—**TODO:**

I. INTRODUCTION

a) Overview: As the demand for cloud services continues to grow, there is an urgent need to develop strategies that minimize the environmental impact of data centers. Evaluating and optimizing the environmental footprint of cloud computing requires a comprehensive understanding of its impacts across multiple dimensions, including carbon emissions, water usage, and land use. The orchestration of workloads in cloud computing environments presents an opportunity to optimize resource utilization and minimize environmental impacts. By leveraging the flexibility of cloud workloads, it is possible to shift computing tasks across time and geography to align with periods and locations of lower environmental impact. This paper explores the potential of such spatio-temporal optimization strategies to reduce the carbon, water, and land use footprints of cloud computing.

b) Motivation: The sustainability of data centers is a rising concern due to their substantial energy consumption and associated environmental impacts. Their energy use is projected to increase significantly in the coming years, driven by the proliferation of data-intensive applications and services, their consumption is expected to reach around 945 TWh globally by 2030¹. Sustainability reports and initiatives are increasingly common and international organizations are setting ambitious targets to reduce the carbon footprint of their operations [1]. Moreover, sustainability in cloud computing

aligns with Agenda 2030 and the Sustainable Development Goals (SDGs), particularly those related to climate action and responsible consumption and production². By addressing the environmental impacts of cloud computing, we can contribute to these global objectives and promote a more sustainable digital future.

While carbon emissions have been the primary focus of sustainability efforts, however, water and land use are also critical factors that must be considered. Recent studies have highlighted the significant environmental impacts of cloud computing beyond carbon emissions. For instance, data centers consume vast amounts of water for cooling purposes, which can strain local water resources [2], [3]. Additionally, large-scale land use, such as one associated with data center construction, can lead to habitat disruption and biodiversity loss [4]. These factors underscore the need for a multi-dimensional approach to evaluating and optimizing the environmental footprint of cloud computing.

c) Contributions: In summary, our contributions are as follows:

- a formulation that optimizes carbon, water and land use impacts of cloud computing, leveraging temporal and/or spatial shifting approaches. To the best of our knowledge, this is the first work to add land use to the evaluation of environmental impacts in cloud computing.
- a comprehensive empirical evaluation using real-world data, quantifying the trade-offs between carbon, water, and land use under various workload and grid scenarios.
- diverse workload types are considered, including big data batch jobs and function as a service (FaaS), to assess the generality of our approach.
- sensitivity analysis of our approach to key parameters such as workload flexibility, grid mix composition prediction accuracy, and seasonality.
- all code and data used in this study are made publicly available to facilitate reproducibility and further research in this area.

d) Paper Structure: The rest of the paper is organized as follows. Section II provides background on data center environmental impacts and workload flexibility. Section III presents our system model and problem formulation. It also describes our methodology, including the proposed optimization algorithms. Section IV presents our simulation setup, including a detailed description of the scenarios characteristics

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¹IEA - Energy demand from AI [Online]

such as workload types, data center specifications, and grid datasets. Section V presents the results of our empirical evaluation and discusses interesting insights and implications of our findings. Section VI reviews related work in carbon-aware computing and sustainability in cloud computing. Finally, Section VII concludes the paper and outlines directions for future research.

II. BACKGROUND

A. Data Centers impacts

a) Operation: Data centers are among the most energy-demanding facilities globally. They require large amounts of electricity to power servers, cooling systems, and backup power supplies. It is estimated that data centers in 2022 accounted for nearly 500 TWh, and it is expected to rise in the coming years [5]. This energy usage results in significant GreenHouse Gas (GHG) emissions, especially in regions where electricity is generated from fossil fuels. In fact, the environmental impact of data centers is directly tied to the energy mix of their respective grids. The energy mix directly influences the **Carbon Intensity (CI)**, which measures the greenhouse gas emissions per unit of electricity consumed (typically expressed in $\text{gCO}_2\text{e/kWh}$); the **Energy Water Intensity Factor (EWIF)**, which captures the amount of water withdrawn or consumed during electricity generation (measured in liters/kWh), which varies with the cooling technologies; and the **Energy Land Intensity Factor (WLIF)**, which quantifies the land use required for producing electricity (expressed in m_2/kWh). An example of these intensities' estimates are reported in [6]–[8]. These are also the main sources used in our experiments (see table VI), and in the discussion section we will analyze how these coefficients relates to each other.

However, the grid energy mix is not the only factor affecting the environmental impact of data centers. The efficiency of data center operations also depends on the infrastructure and technologies used. Introduced by The Green Grid [9], the **Power Usage Effectiveness (PUE)**, is defined as the ratio of the total energy consumed by a datacenter facility to the energy consumed by its IT equipment. A PUE of 1.0 indicates perfect efficiency, where all energy is used exclusively for computation.

Moreover, the efficiency of cooling systems, which are essential to prevent overheating of servers, can significantly affect water usage. The choice of cooling technology (e.g., air cooling vs. liquid cooling) can lead to different water consumption. The **Water Usage Effectiveness (WUE)** (again proposed by The Green Grid [10]), measures the water consumption of a datacenter relative to its IT energy usage, typically expressed in liters of water per kilowatt-hour (l/kWh). Cooling systems have a huge impact on the environment since they are both energy-intensive and water-intensive. Indeed, data centers rely on vast amounts of water for cooling purposes. It is estimated that a typical mid-sized data center can use approximately 25.5 million liters of water each year when employing traditional cooling methods [11]. This heavy water reliance can exacerbate local water scarcity, especially in regions prone to drought.

Finally, since these infrastructures occupy a significant amount of land, the **Land Usage Effectiveness (LUE)**³ should be accounted as well. To the best of our knowledge, this metric has not been standardized and it has not been used previously. Similarly to PUE and WUE, the LUE metric aims to quantify the physical land footprint of data centers in relation to IT energy consumption, capturing the spatial resource requirements of large-scale infrastructure. Thus, for the scope of our work, we define it as the ratio of the total land occupied by the data center property and the IT energy consumption, and thus it is expressed in square meters per kilowatt-hour (m^2/kWh).

b) Construction and Dismantling: Data center facilities also have a significant environmental impact for what concerns the construction and eventual dismantling. Large amounts of raw materials like concrete, steel, copper, and rare earth elements are required for their construction. Moreover, these facilities also occupy significant land. The level of the ecological footprint of a data center building can be formally certified, e.g. with the LEED certification scheme [12]. Moreover, in the context of data centers, it is important to consider not only building materials but also the IT equipment such as servers, networking devices, and storage systems. The production and end-of-life disposal of such hardware contributes significantly to environmental impact [13]. Furthermore, the dismantling of data centers generates substantial e-waste, which often contains toxic materials, like lead, nickel, and mercury, which, if not properly recycled, these can leach into soil and water, ultimately having an adverse impact on human health and the environment [14].

B. Workload flexibility

Workload flexibility refers to the ability to shift workloads in time and space to optimize resource use and minimize environmental impacts. This can involve techniques such as load balancing, resource allocation, and scheduling to take advantage of periods of low demand or renewable energy availability. By increasing workload flexibility, data centers can better align their operations with sustainability goals and reduce their overall environmental footprint.

III. PROBLEM FORMULATION

A. System Model

First of all, let us consider the temporal horizon H , and let us discretize the time. Thus, let us define $\mathcal{T} = [t_0, t_0 + \delta_t, t_0 + 2\delta_t, \dots, H]$, as the set of time steps. Let us define also $\mathcal{D}$ as the set of regions/datacenters, and consider a data center d and a time t . To evaluate its impact accounting for multiple factors (i.e. carbon, water and land use), we define its sustainability profile as follow:

$$\forall t \in \mathcal{T}, d \in \mathcal{D} \quad \mathbf{p}_d(t) = \begin{pmatrix} \mathcal{C}_d(t) \\ \mathcal{W}_d(t) \\ \mathcal{L}_d(t) \end{pmatrix} \quad (1)$$

where:

³K. Sumrani - Decarbonization: From Data Centers to Sustainable Infrastructure

TABLE I
NOTATION

Symbol	Description
$\mathcal{T}$	period discretization
t_0	period start
H	period end
δ_t	step duration
$\mathcal{D}$	set of regions/datacenters
d	region/datacenter
PUE_d	region PUE
WUE_d	region WUE
LUE_d	region LUE
A_d	region land occupation area
IT_d	region IT energy usage
$\mathcal{C}_d(t)$	region carbon footprint
$\mathcal{W}_d(t)$	region water footprint
$\mathcal{L}_d(t)$	region land footprint
$CI_{G_d}^e(t)$	region Energy Carbon Intensity prediction
$EWIF_{G_d}^e(t)$	region Energy Water Intensity Factor prediction
$ELIF_{G_d}^e(t)$	region Energy Land Intensity Factor prediction
$p_d(t)$	data center footprint profile at time t
$p_d^{norm}(t)$	normalized data center footprint profile at time t
$p_{global}^{norm}(t)$	global footprint profile
$\mathcal{V}$	set of VM instances
v	VM instance
CPU_v	number of CPUs
MEM_v	RAM size
$P_{v,j}$	VM v power draw for job j
$P_{v,j}^{CPU}$	VM's CPU power draw based on job CPU utilization
$Watts_{min}$	minimum CPU power draw (fixed for one scenario)
$Watts_{max}$	maximum CPU power draw (fixed for one scenario)
P_v^{MEM}	VM's RAM power draw
B_v	VM's network bandwidth
$\mathcal{J}$	set of jobs
j	job
o_j	job's arrival location
$\mathcal{D}[j]$	job's data availability
s_j	job's data size
n_j	job's number of computing nodes
u_j	job's mean utilization
v_j	job's VM instance
r_j	job's expected runtime
a_j	job's arrival time
t_j	job's deadline time
$\bar{C}(j, d, t)$	scheduling cost (in terms of combination of footprints)
$M(j, d, t)$	migration/data transfer footprints
NEI	Network Energy Intensity
$L(j, d)$	migration/data transfer latency
$E(j, d, t)$	execution footprint

- $\mathcal{C}_d(t)$ is the predicted carbon intensity of datacenter d from time t to time $t + \delta_t$ and it is defined as

$$CI_{G_d}^e(t) * PUE_d \quad (2)$$

It is measured in $\frac{gCO_2}{kWh}$ and represents the carbon emitted per kWh consumed. We consider the carbon intensity prediction of the regional grid of data center d at time t , namely $CI_{G_d}^e(t)$, where e is the predictive model. This intensity is multiplied by the Power Usage Effectiveness of the data center, WUE_d , to account for the overall energy usage of a datacenter for each kWh of computation.

- $\mathcal{W}_d(t)$ is the predicted water intensity and it is defined as

$$WUE_d + EWIF_{G_d}^e(t) * PUE_d \quad (3)$$

It is measured in $\frac{l}{kWh}$ and represents the water consumed per kWh considering the average Energy Water Intensity

Factor prediction of the regional grid, $EWIF_{G_d}^e(t)$ and the Water Usage Effectiveness of the data center, WUE_d .

- $\mathcal{L}_d(t)$ is the predicted land use intensity and it is defined as

$$LUE_d + ELIF_{G_d}^e(t) * PUE_d \quad (4)$$

It is measured in $\frac{m^2}{kWh}$, it is the amount of land used per kWh considering the Energy Land Intensity Factor prediction of the regional grid at time t , $ELIF_{G_d}^e(t)$ and the Land Usage Effectiveness, defined as $LUE_d = \frac{A_d}{P_d^{IT}}$, where A_d is the total area of land occupied by the datacenter property and P_d^{IT} is the energy used to power the IT equipment.

The aforementioned energy intensity factors of the electric grid intensity depends on the energy mix, which is both time and space dependent. Thus, let us define S as the list of sources in the energy mix, namely $\{Solar, Wind, Hydro, Geothermal, Biomass, Nuclear, Coal, Gas, Oil, Unknown\}$, and let $MIX_{G_d,s}^e(t)$ be the share of source $s \in S$ in the mix prediction of G_d . Thus, we can define the energy intensity factors as:

- $CI_{G_d}^e(t) = \sum_{s \in S} MIX_{G_d,s}^e(t) * CI_s$
- $EWIF_{G_d}^e(t) = \sum_{s \in S} MIX_{G_d,s}^e(t) * EWIF_s$
- $ELIF_{G_d}^e(t) = \sum_{s \in S} MIX_{G_d,s}^e(t) * ELIF_s$

where CI_s , $EWIF_s$ and $ELIF_s$ are specified in table VI.

Moreover, let us define $\mathcal{V}$ as the set of virtual machine instances, for each $v \in \mathcal{V}$ it is known the power draw P_v defined as:

$$P_{v,j} = CPU_v * P_{v,j}^{CPU} + RAM_v * P^{MEM} \quad (5)$$

where: CPU_v is the number of CPUs, RAM_v is the size of RAM expressed in GB, B_v is the network bandwidth expressed in Gbps, P^{MEM} is the memory power drawn and $P_{v,j}^{CPU} = Watts_{min} + u_j * (Watts_{max} - Watts_{min})$ is the CPU power draw, both defined according to the cloud carbon footprint methodology [15].

Finally, let us define also the set of job requests J . For each request $j \in J$ it is known: the arrival/origin region/location $o_j \in \mathcal{D}$, the regions in which data is available $\mathcal{D}[j] \subseteq \mathcal{D}$, data size $s_j \in \mathbb{N}$, number of nodes $n_j \in \mathbb{N}$, the expected utilization $u_j \in \mathbb{N}$, the VM instance $v_j \in \mathcal{V}$, the expected runtime $r_j \in \mathbb{Z}$, the arrival time $a_j \in \mathcal{T}$, and the deadline $t_j \in \mathcal{T}$.

B. Evaluating environmental footprints

Firstly, we need a methodology to evaluate the footprints of executing a request in a certain region at a certain time. Let us consider we want to predict the associated footprint of computing a job request j in region/datacenter d at time t . If the data related to the request is not available in d , data migration is required, and it is necessary to account for its environmental impact.

To account for migration footprint overhead, data transmission energy is estimated as follows:

$$M(j, d, t) = \begin{cases} s_j * NEI & \text{if } d \notin \mathcal{D}[j] \\ 0 & \text{otherwise} \end{cases} \quad (6)$$

where NEI kWh/GB is the Network Energy Intensity, as proposed in [16].

To account for the execution footprint we need to know at what time its computation will start. We assume that if data is not migrated the execution start would be at arrival time, since the scheduling execution time is negligible. If the data needs to be migrated we need to compute the data migration latency as follow:

$$L(j, d) = \begin{cases} \frac{s_j}{B_{vj}} & \text{if } d \notin \mathcal{D}[j] \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

To account for the execution footprint, we define

$$E(j, d, t_i) = \int_{t_i + L(j, d)}^{t_i + L(j, d) + r_j} P_j \cdot \mathbf{p}_d^{norm}(t) dt, \quad (8)$$

where $\mathbf{p}_d^{norm}(t) \in \mathbb{R}^3$ is the normalized profile vector of region d for carbon, water, and land use. We define the normalization operator applied componentwise.

$$\mathcal{N}(\mathbf{p}_d(t)) = \frac{\mathbf{x} - \min_{d,t} \mathbf{p}_d(t)}{\max_{d,t} \mathbf{p}_d(t) - \min_{d,t} \mathbf{p}_d(t)}$$

Thus the regional normalized profile is $\mathbf{p}_d^{norm}(t) = \mathcal{N}(\mathbf{p}_d(t))$. We shall consider global intensities to account for migration footprint, and we define them as the normalized (over time) average intensities, namely: $\mathbf{p}_{global}^{norm}(t) = \mathcal{N}(\text{avg}_d \mathbf{p}_d(t))$. Finally, we can compute the overall environmental cost of scheduling the job.

$$C(j, d, t_i) = (E(j, d, t_i) + M(j, n, t_i) * \mathbf{p}_{global}^{norm}(t)) \times \theta \quad (9)$$

where θ is the weight vector to linearly combine the three impact factors, namely: $\theta = (w_C \quad w_W \quad w_L)$.

C. Scheduling Methods

- **Local:** jobs are executed where the request originates.

$$d^*, t^* = o_j, a_j \quad (10)$$

- **Spatial Shifting:** jobs should be executed where their overall impact is minimized (considering data migration when required).

$$d^*, t^* = \text{argmin}_{d \in \mathcal{D}} C(j, d, a_j), a_j \quad (11)$$

- **Temporal Shifting:** jobs should be executed when their overall impact is minimized.

$$d^*, t^* = o_j, \text{argmin}_{t \in [a_t, d_t - r_t, \delta_t]} C(j, o_j, t) \quad (12)$$

- **Spatio-Temporal Shifting:** optimize both when and where the jobs' overall impact is minimized.

$$d^*, t^* = \text{argmin}_{d \in \mathcal{D}, t \in [a_t, d_t - (r_t + L(j, d)), \delta_t]} C(j, d, t) \quad (13)$$

We assume that once data is transferred to another region, the origin location persistently store the data. This is because for periodic jobs we leverage the fact that data is available in multiple regions without transferring it multiple times.

IV. SIMULATION

To evaluate our approach, we conduct two case studies. The first scenario aims to assess the environmental benefit of spatial shifting considering Azure FaaS, while the second considers big data batch jobs deployed in a AWS environment, and aims to assess the environmental benefit of all the approaches proposed. *We leverage real data aggregating several sources, e.g. the energy source mix, the data center-specific environmental parameters. However, some data has been partially randomly generated, e.g. the users' requests.* Although, based on real execution traces, some requests parameters have been randomly generated, e.g. the arrival location or the arrival time (only for AWS).

A. Pseudo-code

Algorithm 1 details the simulation procedure.

Algorithm 1 Simulation Procedure

```

1: Input:  $\mathcal{D}$  data centers, scheduling_method()  $\in \{(10), (11), (12), (13)\}$ 
2: for  $t \in \mathcal{T}$  do
3:   Get sustainability profiles  $p_d(t)$  for each  $d \in \mathcal{D}$ 
4:   Get jobs ready to be scheduled at time  $t$ ,  $J(t)$ .
5:   for  $j$  in  $J(t)$  parallel do
6:      $d^*, t^* = \text{scheduling\_method}()$ 
7:     if scheduling_method()  $\in \{(11), (13)\}$  then
8:        $\mathcal{D}[j] = \mathcal{D}[j] \cup \{d^*\}$ 
9:     end if
10:   end for
11: end for
12: Output: Final schedule over horizon  $H$ 

```

B. Experimental Setup

a) *Hardware and Software Environment:* The experiments were conducted on the following system:

- **CPU:** Intel(R) Core(TM) i9-10920X CPU @ 3.50GHz
- **RAM:** 260 GB
- **Operating System:** Ubuntu 24.04.3 LTS
- **Simulation Framework:** Custom, implemented in python⁴

b) *Scenarios:* The first scenario is based on Azure regions and a FaaS workload. The regions selected for this scenario are "swedencentral", "southcentralus", "centralus" and "eastus". Table II reports regions characteristics. PUEs and WUEs values were taken from Azure issued sustainability reports [17]–[20]. The land occupation data is taken from online available sources (swedencentral⁵, southcentralus⁶, centralus⁷, eastus2⁸). As for the workload, we chose the

⁴GitHub repo

⁵Invest Stockholm, "Two municipalities in the Stockholm region sell land to Microsoft," [Online].

⁶Datacenters.com, "Microsoft Azure: South Central US-Texas," [Online].

⁷Datacenters.com, "Microsoft Azure: Central US-Iowa," [Online].

⁸Datacenters.com, "Microsoft Azure: East US–Virginia (Boydton, VA) Data Center," [Online].

2019 Azure Functions dataset⁹. The source data is composed of two sets of daily CSV files: one detailing per-minute invocation counts for each function, and another providing daily summary statistics of function execution times, including averages and key percentiles (0th, 1st, 25th, 50th, 75th, 99th, and 100th). We utilised the first seven days of this dataset to synthesise 100,000 requests of function execution a day. The selection of a function for each request was probabilistic and weighted by the function's total daily invocation count, such that more frequently invoked functions appeared more often in the trace, mirroring real-world usage patterns. Upon selecting a function, we sampled a synthetic runtime by performing linear interpolation on its known daily execution time percentiles. The arrival time for the request, resolved to the minute-of-day, was similarly sampled from the function's empirical per-minute invocation distribution. Finally, each request was assigned a datacenter location from a predefined set. A single virtual machine configuration (2 vCPU and 4 GB of RAM) is considered for this scenario, as specified in the paper [21].

For this scenario we compare only the **Local** baseline (L) and the **Spatial Shifting** (SP) approach. We assume that the functions are tolerant to the latency required to change regions but not to be delayed in time. Thus, for this reason, we exclude **Temporal Shifting** (T) and **Spatio-temporal Shifting** (STP) scheduling methods. Thus, deadlines are defined for this scenario.

TABLE II
AZURE DATA BY REGION SCENARIO 1.

Region	Location	PUE	WUE	Land Occupation (sqm)
swedencentral	Sweden	1.172	0.16	1300000
southcentralus	Texas	1.307	1.82	43664
centralus	Iowa	1.16	0.19	37904
eastus2	Virginia	1.144	0.17	102193

The second scenario is based on AWS and Big Data workloads. The regions selected for this scenario are: "us-east-1", "us-west-1", "eu-central-1", "eu-west-2", and "eu-north-1". Table IV reports regions characteristics. PUEs and WUEs values were taken from AWS website¹⁰. The land occupation data is estimated based on the 10-k form released by AWS in 2024 [22], in which it is stated that 454546 sqm were owned/rented by AWS for datacenters. The estimation is also based on declared availability zones for each region. Namely, estimated land occupation for a given region is calculated by taking the number of availability zones (AZs) in that region, dividing it by the total number of AZs (117), and then multiplying by the total land occupation of all data centers. As for the workload, we synthetically generate requests from available experimental Spark traces¹¹ [23]. We randomly generated 50k requests over a week. For the arrival pattern, we generated 50% of requests as periodic (in line with literature analysis: about 43% [24], about 60% [25]) and the rest as ad-hoc jobs. We generate synthetic requests for each day of the selected week. The arrival of ad-hoc jobs follows a Poisson distribution

TABLE III
AWS VM INSTANCE TYPES AND SPECIFICATIONS

Type	vCPU	Mem (GB)	Bandwidth (Mbps)
c4.large	2	4	500
c4.xlarge	4	8	750
c4.xxlarge	8	16	1000
m4.large	2	8	450
m4.xlarge	4	18	750
m4.xxlarge	8	34	1000
r4.large	2	16	450
r4.xlarge	4	32	850
r4.xxlarge	8	65	1700

with $\lambda = target_{np}/n_{days} * minutesperday \approx 2.5$. The periodic jobs first instance arrival time is randomly picked within the first 12 hours and their periodicity is set uniformly at random among $\{2, 4, 8, 12\}$ hours. For each request, the origin location is picked uniformly at random from the regions considered for this scenario. The number of nodes and the VM instance and the expected runtime are the ones reported in the traces. Table III summarizes the VM instances considered in our experiments and their specifications are set accordingly (memory, cpu¹², and bandwidth¹³). For this scenario, we consider all of the scheduling methods defined above. We also introduce a variation of **Spatial Shifting**, in which we do not persistently store data transferred when migrating (S). For **Temporal Shifting** (T) and **Spatio-Temporal Shifting** (TSP), the deadline is either its periodicity or, for ad hoc jobs, a tunable delay tolerance parameter.

TABLE IV
AWS DATA BY REGION SCENARIO 2.

Region	Location	PUE	WUE	Land Occupation (sqm)
us-east-1	Virginia	1.15	0.12	233101
us-west-1	California	1.17	0.51	116550
eu-central-1	Frankfurt	1.35	0.01	116550
eu-west-2	London	1.11	0.04	116550
eu-north-1	Stockholm	1.10	0.02	116550

c) *Grid data and intensities*: We collected data from various sources. Since we considered multiple regions, we collected publicly available data from different grid operators (sw¹⁴, uk¹⁵, de¹⁶, ercot¹⁷, miso¹⁸, pjm¹⁹, caiso²⁰). The historical mix data is used to compute the actual environmental impact of the workload execution, while to make decisions noised mix data is used to simulate the prediction of grids' mix. The data sources are not standardized, thus we had to harmonize the data to have a common format. We considered a granularity of 1 hour and the data with a finer grain were reshaped considering the mean value for each hour. Table V depicts the relationship between grids and cloud providers' regions.

¹²Amazon EC2 Instance types [Online]

¹³Specifications for Amazon EC2 previous generation instances [Online]

¹⁴ENTSOE Transparency Platform [Online]

¹⁵National Electricity System Operator API [Online]

¹⁶SMARD - Download market data [Online]

¹⁷EIA - Wholesale Electricity Market Data - ERCOT [Online]

¹⁸EIA - Wholesale Electricity Market Data - MISO[Online]

¹⁹EIA - Wholesale Electricity Market Data - PJM [Online]

²⁰EIA - Wholesale Electricity Market Data - CAISO [Online]

⁹Github: Azure Functions Trace 2019 [Online]

¹⁰AWS Cloud: Increasing Efficiency [Online]

¹¹Github: scout: A Tool for Cloud Configuration Optimization [Online]

TABLE V
DATA SOURCES FOR GRID MIXES

Grid abbreviation	Regions
sw	eu-north-1, swedencentral
uk	eu-west-2
de	eu-central-1
ercot	southcentralus
miso	centraus
pjm	us-east-1, eastus2
caiso	us-west-1

To incorporate uncertainties into the grid mix data, we introduce synthetic noise. Specifically, for each time step and region, the renewable share is perturbed by an error term ϵ_r , sampled from a normal distribution $\mathcal{N}(0, \sigma_r)$, where $\sigma_r = mae \cdot \frac{1}{\sqrt{2/\pi}}$ and mae is the target mean absolute error. The error ϵ_r is then distributed among the renewable sources (solar, wind, geothermal, hydro, biomass) according to a probability distribution designed to reflect their prediction difficulty: solar gets 45% of error, since it is the hardest source to estimate [26]; wind gets 30% [27]; both hydro [28] and geothermal [29] get 10%; and biomass get 5% of the error, being the most controllable among the renewable sources. Each renewable source i receives a noise $\epsilon_i = \epsilon_r \cdot w_i$, where w_i is the weight for source i . This approach is a tentative to simulate realistic prediction errors and reflects the varying uncertainty associated with each energy source. For non-renewable sources, a uniform noise is applied to each share to maintain the total mix balance.

Finally, in our simulations, to compute the average intensities of the mixes, we consider the intensity coefficients reported in Table VI. Carbon intensity values correspond median of the lifecycle emissions reported in IPCC Annex III, Table IPCC A.III.2 table (page 1335) [6]. Except for the oil carbon intensity, which is missing in such source, then, we consider the coefficient provided by [30]. Water intensity values were obtained from the Water Consumption Factors presented in Tables 1 and 2 (pages 12-13) of the NREL technical report on energy-related water consumption [7]. Land use intensity values were extracted from a comprehensive survey [8]. The survey does not provide a value for oil, but the authors assume it to be comparable to gas, so we did as well. For all the factors, when multiple options were presented for the same energy source, we considered the average of median intensity values. Note that we could not find any source for the oil EWIF, thus we assume the mean of the other sources' intensity values. As well as for the unknown share.

d) Other parameters: For both scenarios we considered one-week horizon and varied the week in different seasons, namely: winter ($t_0 = 2024-01-15T00:00:00Z$, $H = 2024-01-22T00:00:00Z$), spring ($t_0 = 2024-04-15T00:00:00Z$, $H = 2024-04-22T00:00:00Z$), summer ($t_0 = 2024-07-15T00:00:00Z$, $H = 2024-07-22T00:00:00Z$), autumn ($t_0 = 2024-10-15T00:00:00Z$, $H = 2024-10-22T00:00:00Z$). And the step δ_t is set to one minute. To account for the uncertainty in the prediction of the workload, we considered different values of MAE, namely: 0.05, 0.1, 0.15, 0.2. For statistical significance, each simulation is repeated 6 times with different random

TABLE VI
IMPACT COEFFICIENTS FOR VARIOUS ELECTRICITY GENERATION TECHNOLOGIES. **SOURCES:** CARBON: IPCC [6], OUR WORLD IN DATA [30] (OIL). WATER: NREL [7]. LAND USE [31].

Energy Source	CI gCO_2e/kWh	EWIF l/kWh	ELIF ha/TWh
Nuclear	12	1.957	7.1
Geothermal	38	5.827	45
Biomass	485	1.145	29065
Coal	820	1.8	1000
Wind	11.5	0	6065
Solar	38.67	1.385	1650
Hydro	24	17	650
Gas	490	1.099	1155
Oil	720	n/a (3.776)	1155
Unknown	n/a (293.24)	n/a (3.776)	n/a (4532)

seeds, namely: 0, 1, 2, 3, 4, 5. The factors weights are set to: [1.0, 0.0, 0.0] only carbon footprint is optimized, [0.0, 1.0, 0.0] only water footprint is optimized, [0.0, 0.0, 1.0] only land footprint is optimized, [0.333, 0.333, 0.334] all footprints are optimized equally. Moreover, to account for the energy consumption of the network, we considered a network energy intensity of 0.06 kWh/GB [32]. The memory power draw P^{MEM} , and the minimum and maximum CPU power draw ($Watts_{min}, Watts_{max}$) are respectively set to 0.357 Watt/GB and (0.74, 3.5) for AWS and (0.78, 3.76) for Azure [15]. Finally, we set P_d^{IT} to the estimate provided by EIA. Assuming the IT power draw to be 100 MW we assumed an IT power consumption of $8760 * 100 * 10^3$ kWh over one year.

V. RESULTS AND DISCUSSION

In this section, we present and discuss the results of our experiments under the two considered scenarios. For each scenario, we evaluate different shifting strategies, and assess their sensitivity to grid's renewable share estimation errors as well as seasonal variations.

A. Energy Intensity Factors

To understand the relationship between the carbon intensity and the other environmental factors, we plot the energy intensity factors normalized for each energy source. It seems that carbon and water are negatively correlated (e.g. hydro, coal), but not all low carbon intensity sources have high water intensity (e.g., wind and nuclear). On the other hand, land use seems to not be correlated with the other factors, although there are sources with high land use and low water intensity (e.g. biomass), and sources with low land use and high carbon and water intensity (e.g. oil).

B. Scenario 1: Azure + FaaS

For this first scenario we analysing the benefits of spatial shifting optimizing for all factor weight combinations.

a) Spatial shifting: Figure 2 shows the footprint improvement with respect to the local baseline of spatial shifting. This experiment considered the winter week and an error of 10% on the renewable share prediction. We compare all factor weights' combinations defined above. The results

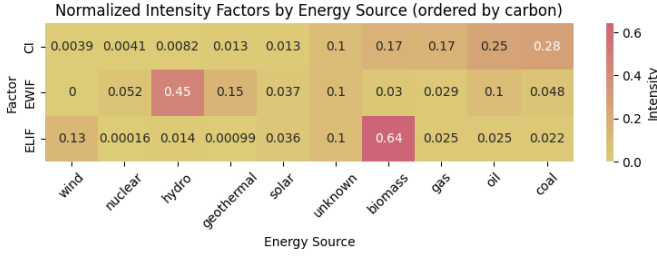


Fig. 1. Energy Intensity Factors normalized

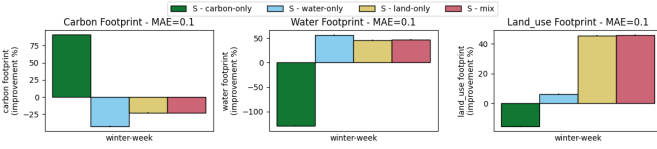


Fig. 2. Spatial shifting - Azure - percent improvement over local baseline - Winter week, mae = 0.1

demonstrate that regional shifting of FaaS workloads across Azure regions leads to noticeable improvements in carbon efficiency. The plots show that when optimising for a particular metric (carbon, water, or land use footprint), we can achieve a 45 – 85% improvement in that metric. When considering the metric for which the optimisation was performed, we see that the carbon footprint benefits the most, with up to an 85% reduction in emissions compared to the local baseline. This significant reduction reflects the zero migration cost of FaaS workloads. As expected, optimising for one metric can lead to trade-offs in the other metrics. For instance, optimising for carbon footprint lead to a significant increase in water footprint, as already observed in previous studies [3]. For instance, a reduction of about 85% in carbon footprint leads to an increase of about 40% in water footprint. Also the land use footprint increases, but to a lesser extent, about 25% increase. Similarly, optimizing for water footprint (about 50% reduction) leads to a significant increase in carbon footprint (about 125% increase), but the land use footprint decreases proportionally to the water footprint reduction (about 50% reduction). On the other hand optimizing for land use footprint (about 45% reduction) leads to a moderate increase in carbon footprint (about 15% increase) and a small reduction in water footprint (less than 10% reduction).

Figure 4 shows the request distribution across the regions. To help us understand this distribution, let us consider Figure 3, which simultaneously reports intensity values over the winter week and the efficiency metrics for each factor for each region. The "swedencentral" region results to be the most used when optimizing for carbon footprint, while "centralus"

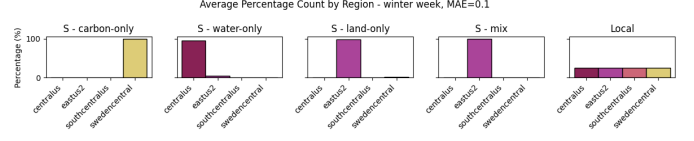


Fig. 4. Spatial shifting - Azure - Request distribution across different regions for different criteria - Winter week, mae = 0.1

is the most used when optimizing for water footprint with a small percentage of requests going to eastus2, which in turns is the most used region for land use footprint with a small percentage of requests going to "swedencentral" as well. This indicates that "swedencentral" is the optimal region for minimizing carbon footprint, given its grid carbon intensity, even if the region has not the smallest PUE value (1.172). Similarly, "centralus" is the optimal region for minimizing water footprint even if has not the smallest WUE (0.19) nor the one with the smallest grid water intensity. The region "southcentralus" would be the one with the smallest grid intensity, however it is also the one with the worst WUE. On the other hand, "swedencentral" region would be the most water efficient in terms of WUE, however it is the region with the worst grid water intensity. Finally, "eastus2" results to be the optimal region for minimizing land use footprint since, even if it is not the region with less land occupation, it is the one which grid has sensible lower land use intensity. This suggests that the energy mix has a more significant impact on the environmental metrics than the data center efficiency metrics.

Overall, these results highlight the potential of spatial shifting significantly enhancing the sustainability of FaaS workloads by leveraging regional variations in grid intensities.

b) Grid error sensitivity analysis: Table ?? shows the improvement with respect to the local baseline in the three metrics for different optimisation strategies, under different error levels of the generation mixes. The results indicate that the improvement in reduction in all three metrics remains robust even for higher error levels.

c) Seasonality sensitivity analysis: To understand how seasonality affects the effectiveness of the scheduling methods, we perform a sensitivity analysis considering the four weeks: winter, spring, summer, and autumn. We use a fixed prediction error of mae = 0.1 for this analysis. The analysis shows that the effectiveness of spatial shifting varies with seasonal changes in grid intensities, but the overall trend of improvement remains consistent. Land use factors, in particular, demonstrate varying impacts across different seasons. The highest improvements are observed during the autumn season. In contrast, the summer season shows more moderate improvements.

C. Scenario 2: AWS + Big Data

For this second scenario, not only we investigate the spatial shifting method, but also temporal shifting and spatio-temporal shifting. We consider the same week in winter as in Scenario 1. We consider a prediction error (mae) of 0.1. We consider the same factor weights' combinations as in Scenario 1.

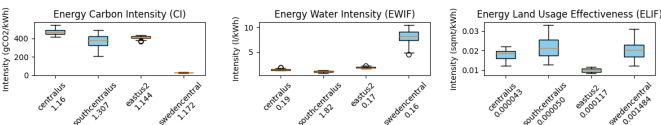


Fig. 3. Grid intensities - Azure - Winter week, mae = 0.1

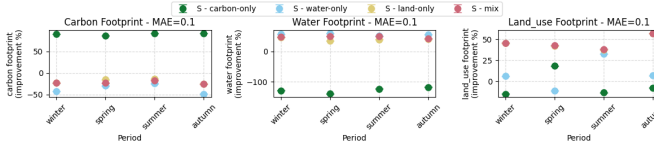


Fig. 5. Spatial shifting - Azure - Percent improvement over local baseline - Seasonality sensitivity analysis, mae = 0.1

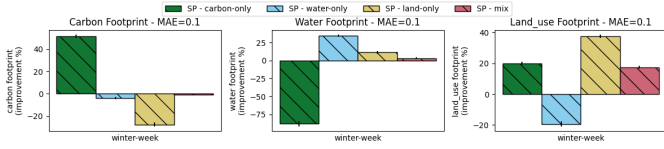


Fig. 6. Spatial shifting - Aws - Percent improvement over local baseline - Winter week, mae = 0.1

For temporal shifting we consider different values of delay tolerance (dt) = {4, 12, 24, 48} hours.

a) *Spatial shifting*: Figure 6 shows the footprint improvement with respect to local baseline of spatial shifting (SP), comparing all factor weights' combinations defined above. Similar to Scenario 1, we observe a significant improvement in each metric when optimising for it. In this case the improvements range from 20 – 45% reduction compared to the local baseline.

Figure 8 show the request distribution across the regions. With respect to Scenario 1, we observe a more balanced distribution across the regions. This is likely due to migration overhead, which is negligible for Faas workloads, but not for big data workloads, thus it limits the number of migrations leading to a more balanced distribution. However, we still observe a significant shift of requests. To help us understand this distribution, let us consider Figure 7, which simultaneously reports intensity values over the winter week and the efficiency metrics for each factor for each region. When optimising for carbon, about 50% of requests are scheduled to "eu-north-1" which is the most energy efficient region (PUE 1.1) and also the region with the grid with lowest carbon intensity. When optimising for water, about 40% of requests are scheduled to "us-west-2" which is not the most water efficient region (WUE 0.04) but it has the grid with lowest impact on water.

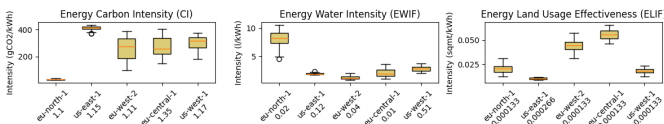


Fig. 7. Grid intensities - Aws - Winter week, mae = 0.1

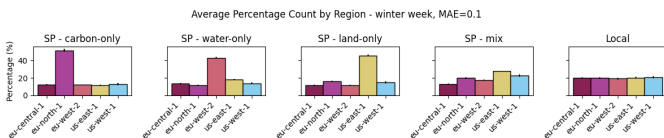


Fig. 8. Spatial shifting - AWS - Request distribution across different regions for different criteria - Winter week, mae = 0.1

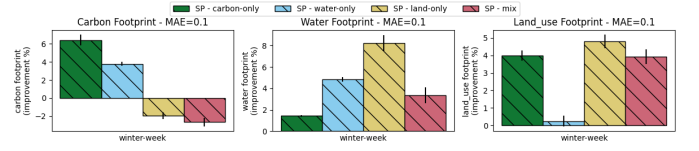


Fig. 9. Spatial shifting (persistent storage vs non-persistent storage) - Aws - Percentage improvement over non-persistent storage of migrated data - Winter week, mae = 0.1

TABLE VII
SPATIAL SHIFTING (PERSISTENT STORAGE VS NON-PERSISTENT STORAGE) - AWS - AVERAGE PERCENTAGE OF MIGRATION FOOTPRINTS - WINTER WEEK, MAE = 0.1

Scheduler / Weights	Carbon (%)	Water (%)	Land Use (%)
L	0	0	0
S - carbon-only	0.13	0.03	0.08
SP - carbon-only	0.07	0.02	0.04
S - water-only	0.03	0.05	0.03
SP - water-only	0.02	0.03	0.02
S - land-use-only	0.04	0.05	0.07
SP - land-use-only	0.02	0.03	0.04
S - mix	0.02	0.02	0.02
SP - mix	0.01	0.01	0.01

Finally, when optimising for land use, about 45% of requests are scheduled to "us-east-1", which even if is the region with the most land occupation, it is the one which grid has the lowest impact on land use. Also for this scenario, the grid mix has a higher impact on the environmental factors than the efficiency of the regions.

Moreover, we consider an alternative approach to spatial shifting (S). In this approach the data associated with each request must be transferred each time a migration occurs, even if the data was transferred to that region in a previously. Figure 9 shows the improvement of SP over this alternative approach S. The plot shows that when optimising for a particular metric (carbon, water, or land use footprint), we can achieve about 5 – 6% improvement in that metric. Table VII provides a summary of the migration overhead for the three factors, it shows the percentage of footprints caused by migration. The migration overhead varies based on the scheduling algorithm and the weights used, and it is relatively low, ranging from 0.01% to 0.13% for carbon, 0.01% to 0.05% for water, and 0.01% to 0.08% for land use. The overhead of SP is about half compared to S. Overall, the migration overhead remains minimal. This indicates that the benefits of spatial shifting in reducing the overall footprint significantly outweigh the footprints associated with migration.

b) *Temporal shifting*: Temporal shifting allows deferring the execution of a workload within a certain delay tolerance (dt) to exploit times of the day when the grid mix is cleaner (resulting in lower intensities). This approach does not involve migrating workloads across regions, thus it does not incur migration overhead. However, it may introduce execution delays, which can impact user experience. However, for big data workloads, we consider that this delay may be acceptable.

Figure 10 shows the footprint improvement with respect to the local baseline of temporal shifting (T) considering increasing values of delay tolerance (dt). As one would expect,

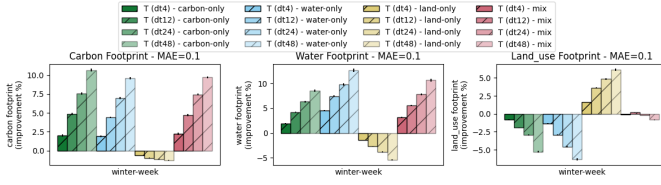


Fig. 10. Temporal shifting - Aws - Percent improvement over local baseline - Winter week, mae = 0.1

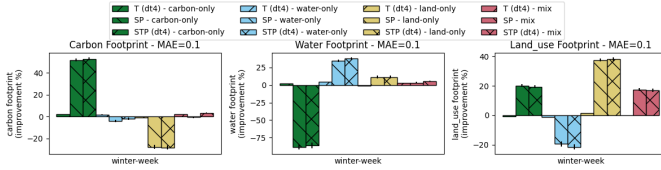


Fig. 11. Spatio-temporal shifting (persistent data storage when migrating) - Aws - Percent improvement over local baseline - Winter week, mae = 0.1

increasing the delay tolerance allows for greater flexibility in scheduling workloads at times when the grid is cleaner, leading to more significant reductions in carbon footprint. However, the improvements are relatively modest compared to spatial shifting, with reductions ranging from 6% to 12% even at the highest delay tolerance of 48 hours. This suggests that while temporal shifting can contribute to sustainability goals, its impact is limited by the inherent variability of renewable energy availability within a single region.

c) *Spatio-temporal shifting*: Combining spatial and temporal shifting can yield greater sustainability benefits by leveraging both regional differences in grid mix and variations in the grid mix over time. Our results confirm this, but the improvement with respect to only spatial shifting is small.

Figure 11 shows the footprint improvement with respect to the local baseline to show the benefit of combining spatial and temporal shifting. However, note that this gain could be larger for larger values of delay tolerance (dt), as previously shown for the temporal shifting method.

d) *Grid error sensitivity analysis*: Also for this scenario, we perform a sensitivity analysis to understand how the prediction error (mae) affects the effectiveness of the spatial shifting method. We consider again only the winter week and the same error values, $mae = \{0.05, 0.1, 0.15, 0.2\}$.

As shown in Figures 12 and 14, spatial shifting (SP) and spatio-temporal shifting (STP) is not affected by fluctuations of the grid mix renewable share prediction error. Conversely, we observe that the effectiveness of temporal shifting decreases as the prediction error increases, as shown in Figure 13. This suggests that, unlike temporal shifting approach, the spatial shifting methods are more robust and can still provide significant sustainability benefits despite uncertainties in grid mix predictions.

e) *Seasonality sensitivity analysis*: Following the same approach used for scenario 1 we conduct seasonality sensitivity analysis for this scenario as well.

Figure 15 shows the footprint improvement with respect to the local baseline for spatial shifting (SP) for each season. We observe that the improvements have small variations across

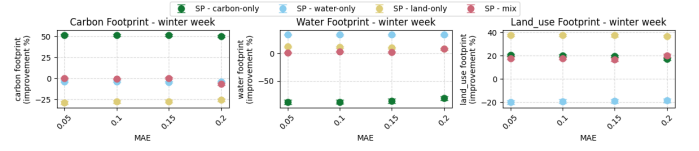


Fig. 12. Spatial shifting (SP) - Aws - Percent improvement over local baseline - winter week - Sensitivity analysis mae

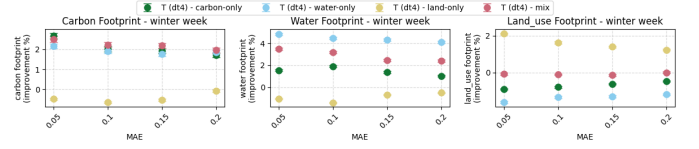


Fig. 13. Temporal shifting (T) - Aws - Percent improvement over local baseline - winter week, dt=4 - Sensitivity analysis mae

seasons. Thus, the effectiveness of spatial shifting appears to be consistent through irrespective of the season. Conversely, temporal shifting seems to be more affected by seasonality. For example, when optimization for carbon footprint in winter we achieve around 2% improvement over the local baseline, while in autumn such improvement reaches 5%, as shown in Figure 16. Finally, Figure 17 shows the footprint improvement over the baseline for spatio-temporal shifting. We observe a behavior similar to that of spatial shifting, confirming the effectiveness of the contribution of spatial shifting to the combined method.

D. Scenario 1 vs Scenario 2

From the results we can see that with diverse providers and workloads, the results are different but the improvements are relevant in both. In particular, we can see that for Azure we obtain the highest improvements, since there is no footprint overhead when migrating, while for AWS there is a small overhead. However, this overhead is such that the improvements are still significant. Moreover, it allows us to obtain a more balanced distribution of requests among the regions.

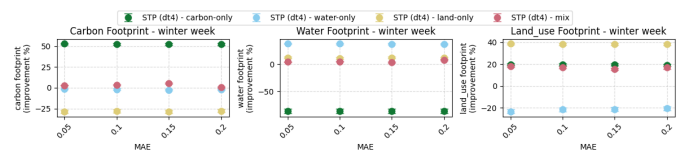


Fig. 14. Spatio-temporal shifting (STP) - Aws - Percent improvement over local baseline - winter week, dt=4 - Sensitivity analysis mae

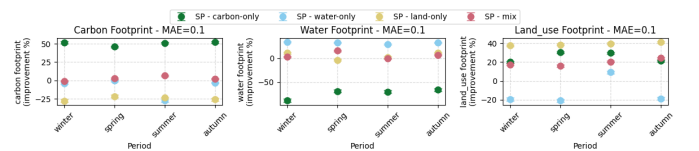


Fig. 15. Spatial shifting (persistent data storage when migrating) - Aws - Percent improvement over local baseline - mae = 0.1 - Sensitivity analysis seasonality

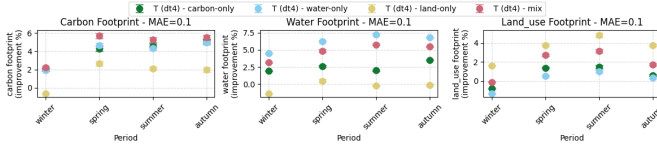


Fig. 16. Temporal shifting - Aws - Percent improvement over local baseline - mae = 0.1 - Sensitivity analysis seasonality

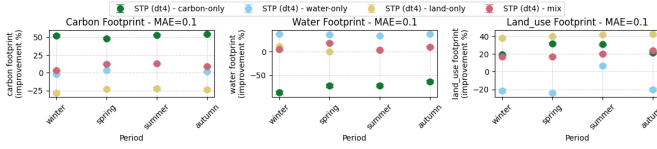


Fig. 17. Spatio-temporal shifting - Aws - Percent improvement over local baseline - mae = 0.1 - Sensitivity analysis seasonality

VI. RELATED WORK

Enhancing resource management and operational efficiency, cloud orchestration has fueled a wide body of research [33]. Allocating resources, both optimizing energy consumption and carbon awareness [34]–[40] have received a significant interest. Recently, the impact on water resources of computation has raised some concerns, but few studies have focused on this issue/aspect [2], [3]. Differently from previous works, we included also land use into an approach that assesses and optimizes the overall impact.

A. Carbon-Aware Computing

In cloud computing, carbon-aware optimisation focuses on aligning workload execution with variations in the carbon intensity of the electricity supply over time and/or across different geographical areas.

a) *Temporal Shifting*: Radovanović et al. [41] present Google’s system for global Carbon-Intelligent Compute Management, which, by delaying temporally flexible workloads, minimises carbon footprint of the electricity used Wiesner et al. [42] analyse the temporal shifting benefits considering different regions, demonstrating the carbon savings potential of delay-tolerant workloads in Germany, Great Britain, France, and California. Piontek et al. [39] propose a Kubernetes scheduler that shifts non-critical jobs in time so to reduce carbon emissions based on their prediction algorithm.

b) *Spatial Shifting*: Gao et al. [35] provide a scheduler that controls the traffic directed to each data center optimizing a 3way-trade-off between access latency, electricity cost, and carbon footprint. Maji et al. [37] propose load balancing in VMware’s Avi Global Server Load Balancer, which uses a linear scoring function to select the optimal data center in terms of marginal carbon intensity and the distance between the client and the data center. Chadha et al. [40] present GreenCourier, a Kubernetes scheduler designed to reduce carbon emissions associated with serverless functions scheduled across geographically distributed regions. Lindberg et al. [43] propose a metric, named locational marginal carbon emission, to guide spatial shifting. Zheng et al. [44] evaluate the idea of migrating workloads between data centers to reduce

curtailment. It has been shown that migrating workloads from fossil-fuel-powered grids to renewable-energy-powered grids during curtailment hours simultaneously reduces emissions and absorbs excess renewable generation.

c) *Spatio-Temporal Shifting*: Zhou et al. [36] use Lya-punov optimization to perform load balancing across geo-distributed data centers, capacity right-sizing (turning off idle servers), and server speed scaling (adjusting the CPU frequency). The objective here is to optimize a 3way-trade-off between electricity cost, SLA (Service-Level Agreement) requirements, and emission reduction budget. Souza et al. [34] propose a provisioner that minimizes both the number of active servers and the associated carbon emission. Sukprasert et al. [45] analyses the benefits and limitations of carbon-aware spatiotemporal shifting for cloud workloads. Cordingly et al. [38] propose a prototype for computing resource aggregation that minimizes the carbon footprint of a serverless application through carbon-aware load distribution.

B. Beyond Carbon-Aware Computing

Recent research has started to address the environmental impact of cloud and AI workloads beyond just carbon emissions. Li et al. [2] highlight the hidden water footprint of AI models, uncovering the significant amounts of freshwater consumed for both cooling and electricity generation. Their study estimates that training large AI models results in hundreds of thousands of liters of direct water consumption. Jiang et al. [3] introduce WaterWise, a job scheduling framework that co-optimizes carbon and water sustainability for geographically distributed data centers by employing a Mixed Integer Linear Programming (MILP)-based scheduler, which dynamically shifts workloads to regions with optimal carbon and water conditions while considering delay tolerance constraints. This work aligns with ours, which also emphasizes the need for multi-dimensional sustainability metrics when optimizing cloud workload allocation, highlighting the need for comprehensive sustainability-aware orchestration strategies.

Although carbon and water footprints have been studied, land use has not been considered in cloud sustainability research. Therefore, the question of how to incorporate land use considerations remains an open research challenge.

VII. CONCLUSION

To conclude, we summarize the key contributions of our study highlighting the obtained results and the limitations and threats to validity of our work.

a) *Lesson learned*: : In this paper we have shown that spatial and temporal shifting can significantly reduce the environmental footprint of cloud workloads. Spatial shifting, in particular, has demonstrated substantial improvements across all three environmental factors considered (carbon, water, and land use), with reductions ranging from 20% to 85% depending on the scenario and optimization criteria. Temporal shifting also contributes to footprint reduction, to a lesser extent, with improvements up to 12%. The combination of both methods yields the highest reductions, although the incremental benefit over spatial shifting alone is small. Our sensitivity analyses

indicate that these methods are robust to prediction errors in grid mix data and maintain their effectiveness across different seasons. Overall, our findings highlight the potential of a more sustainable workload scheduling as a practical approach considering diverse cloud providers and workloads.

b) Limitations: : This study has several limitations and threats to validity that should be considered when interpreting the results. Firstly, the scenarios considered may not fully represent the diversity of real-world cloud environments, as we focused on specific providers (Azure and AWS) and workload types (FaaS and big data). Moreover, the workload traces used for the AWS scenario may not be representative of production cloud workloads since arrival patterns are synthetically randomized. Secondly, the accuracy of our results is contingent upon the quality of the input data, including grid mix predictions that in our case are synthetically generated from the historical data using an arbitrary noise distribution. Moreover, to estimate the LUE parameter, we use estimations for the region IT power consumption and the land occupation, since precise data is available. This, this estimates could affect the reliability of our findings. Thirdly, our models assume that workloads can be freely shifted across regions and time without considering potential constraints such as latency requirements, data sovereignty issues, or user experience impacts. Finally, our simulations do not account for potential economic costs associated with shifting workloads, which could influence the feasibility of implementing such strategies in practice.

c) Future directions: : Future work could explore the integration of additional environmental factors into the simulations (e.g. environmental impacts related to construction and dismantling of datacenters), as well as the introduction of real-world constraints (e.g. capacity budgeting and load balancing). Furthermore, investigating the trade-offs between different optimization criteria (e.g., cost vs. sustainability) could provide valuable insights.

ACKNOWLEDGMENTS

HOWTO: This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this article.

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VIII. BIOGRAPHY SECTION

If you have an EPS/PDF photo (graphicx package needed), extra braces are needed around the contents of the optional argument to biography to prevent the LaTeX parser from getting confused when it sees the complicated `\includegraphics` command within an optional argument. (You can create your own custom macro containing the `\includegraphics` command to make things simpler here.)

If you include a photo:

Michael Shell Use `\begin{IEEEbiography}` and then for the 1st argument use `\includegraphics` to declare and link the author photo. Use the author name as the 3rd argument followed by the biography text.

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